Adult Narrow-Complex Tachycardia

AC-06 | Adult Cardiac

Quick flow

01 Assess in Clinical Context [EMT / AEMT / Paramedic]

Airway/ventilation • oxygen if hypoxemic • monitor/pads • BP/SpO_2 • IV • 12-lead when it will not delay urgent treatment

02 Sinus Tachycardia / Secondary Cause? [EMT / AEMT / Paramedic]

Treat hypoxia, pain, fever, dehydration, hemorrhage, sepsis, stimulant exposure, or another underlying cause

03 Instability Caused by Rhythm? [EMT / AEMT / Paramedic]

Hypotension • acute altered mental status • shock • ischemic chest discomfort • acute heart failure

04 Unstable → Synchronized Cardioversion [Paramedic]

Regular narrow 100 J • AF 200 J • flutter 200 J • sedate when feasible without delaying shock • resynchronize every time

05 Stable + Regular [EMT / AEMT / Paramedic]

Modified Valsalva • no carotid massage • if unsuccessful, request Paramedic adenosine pathway below

06 Adenosine [Paramedic]

Paramedic: 6 mg rapid IV/IO + immediate flush • then 12 mg after 1-2 min if needed; may repeat 12 mg once • continuous rhythm recording

07 Stable AF / Flutter [Paramedic]

No routine adenosine • SBP ≥110 and no AV-nodal-agent contraindication • use diltiazem when available; during approved Cardizem shortage, metoprolol 2.5-5 mg IV push over 1 min every 10 min (max 15 mg)

08 Reassess + Transport [EMT / AEMT / Paramedic]

Post-treatment rhythm strip/12-lead • complete vital signs • monitor recurrence • route wide or polymorphic rhythms appropriately

Overview

- Adult pathway for patients 16 years of age and older with a pulse and a narrow-complex tachycardia, generally defined by QRS duration under 0.12 seconds.
- Determine whether the tachyarrhythmia is causing instability or represents compensation for another problem before attempting rhythm conversion.
- A tachyarrhythmia causing instability is typically at least 150/min, but treatment is determined by the patient's clinical condition rather than rate alone.

EMT Care

Recognition, supportive care, monitoring, pad placement, modified Valsalva, and rapid escalation for instability.

EMT

- Perform the primary assessment; support airway and ventilation, give oxygen only when hypoxemic, and assess whether the tachycardia is likely causing the patient's symptoms.
- Obtain complete vital signs, apply the cardiac monitor/AED and defibrillation pads, and obtain a 12-lead ECG when trained and equipped without delaying urgent treatment.
- Identify and treat likely causes of sinus tachycardia, including hypoxia, pain, fever, dehydration, hemorrhage, sepsis, and stimulant exposure.

- For a stable, regular narrow-complex rhythm, coach a modified Valsalva maneuver. Do not perform carotid massage.
- Immediately request paramedic intervention and prepare for synchronized cardioversion when instability is attributable to the tachycardia.

AEMT Care

All EMT care plus vascular access, ECG acquisition, treatment preparation, and monitoring during cardioversion.

AEMT

- Perform all EMT actions and establish IV/IO access without delaying synchronized cardioversion or transport.
- Acquire a 12-lead ECG before treatment when feasible and assist with continuous rhythm recording during adenosine administration.
- Prepare medication, suction, oxygenation/ventilation equipment, and airway equipment for cardioversion and procedural sedation.
- Reassess blood pressure, mental status, perfusion, chest discomfort, respiratory status, and rhythm after every intervention.
- Do not independently administer adenosine, diltiazem, or procedural sedation under AC-06.

Paramedic Care

All AEMT care plus rhythm interpretation, adenosine, diltiazem or the approved shortage-only metoprolol alternative, synchronized cardioversion, and procedural sedation.

Paramedic

- Perform all EMT and AEMT actions; determine rhythm regularity, QRS duration, hemodynamic effect, and whether the tachycardia is the cause of instability.
- For unstable regular narrow-complex tachycardia, cardiovert at 100 J. For unstable atrial fibrillation or flutter, cardiovert at 200 J.
- Confirm synchronization markers before every shock and resynchronize after every cardioversion attempt; increase energy after an unsuccessful shock.
- Sedate whenever feasible with midazolam 2-5 mg IV/IO using the smallest effective dose, but do not delay cardioversion.
- For stable regular narrow-complex tachycardia after unsuccessful modified Valsalva, give adenosine 6 mg rapid IV/IO with an immediate flush, then 12 mg after 1-2 minutes if needed; may repeat the 12 mg dose once.
- For significantly symptomatic stable atrial fibrillation/flutter with SBP of at least 110 mmHg and no contraindication, give diltiazem 0.25 mg/kg IV/IO (maximum 20 mg); if needed, repeat 0.35 mg/kg IV/IO after 15 minutes (maximum 20 mg).
- During the approved Claiborne Cardizem shortage only, when diltiazem is unavailable and the same AF/flutter eligibility criteria are met, give metoprolol 2.5-5 mg IV push over 1 minute every 10 minutes; maximum three doses or 15 mg. Reassess blood pressure, heart rate, rhythm, perfusion, respiratory status, and symptoms before every dose.
- Do not independently combine or sequence diltiazem and metoprolol. Contact Medical Control for inadequate response, intolerance, or uncertainty; do not use amiodarone as an AV-nodal substitute for stable AF/flutter with RVR.
- Route suspected pre-excited, wide-complex, or polymorphic rhythms to the appropriate pathway and contact Medical Control early.

Indications

- Narrow-complex tachycardia with a pulse in a patient 16 years of age or older.
- Stable regular narrow-complex tachycardia potentially responsive to modified Valsalva or adenosine.

- Atrial fibrillation or atrial flutter with rapid ventricular response requiring assessment for instability or rate control.

Contraindications

- Do not administer adenosine or perform cardioversion for sinus tachycardia; treat the underlying cause.
- Do not administer adenosine for an irregular rhythm, unstable wide-complex tachycardia, polymorphic tachycardia, or active severe bronchospasm.
- Do not administer diltiazem or the shortage-only metoprolol alternative for hypotension, cardiogenic shock, acute decompensated heart failure, suspected systolic dysfunction, acute coronary syndrome, pre-excited atrial fibrillation/flutter, bradycardia, advanced heart block, or an undifferentiated wide-complex rhythm.

Assessment | Initial assessment

- Maintain a patent airway, assist ventilation as needed, and administer oxygen only for hypoxemia.
- Obtain continuous cardiac monitoring, defibrillation pads, frequent blood pressure, SpO_2, IV access, and two complete vital-sign sets when feasible.
- Obtain a 12-lead ECG before treatment when the patient is stable enough and it will not delay cardioversion.
- Determine onset, prior episodes, medications, stimulant or toxicologic exposure, cardiac history, heart-failure history, anticoagulant use, and known pre-excitation.

Assessment | Instability attributable to tachycardia

- Hypotension.
- Acutely altered mental status.
- Signs of shock or poor perfusion.
- Ischemic chest discomfort.
- Acute heart failure.
- Confirm that the rhythm is causing instability rather than representing a compensatory response to another condition such as sepsis, hemorrhage, hypoxia, or pain.

Assessment | Rhythm classification

- Confirm QRS duration under 0.12 seconds and determine whether the rhythm is regular or irregular.
- Regular narrow rhythms may include AVNRT, AVRT, atrial tachycardia, atrial flutter with fixed conduction, or sinus tachycardia.
- Irregular narrow rhythms commonly include atrial fibrillation, atrial flutter with variable conduction, or multifocal atrial tachycardia.
- If the rhythm is wide, irregular with pre-excitation, or polymorphic, stop this pathway and open AC-07 or AC-08.

Treatment steps

- Treat hypoxia, pain, fever, dehydration, hemorrhage, sepsis, stimulant exposure, and other reversible causes. Do not treat sinus tachycardia with adenosine or cardioversion.
- For instability attributable to the tachycardia, perform immediate synchronized cardioversion: 100 J for regular narrow-complex tachycardia and 200 J for atrial fibrillation or atrial flutter.
- Confirm synchronization markers before every shock and resynchronize after every attempt. Increase energy after an unsuccessful synchronized shock.
- If synchronization is delayed and the patient's condition is critical, deliver an unsynchronized high-energy shock.
- Sedate whenever feasible with midazolam 2-5 mg IV/IO using the smallest effective dose, but do not delay cardioversion.

- For stable regular narrow-complex tachycardia, perform a modified Valsalva maneuver. Do not perform carotid massage.
- If modified Valsalva is unsuccessful, give adenosine 6 mg rapid IV/IO push through the most proximal practical access followed immediately by a rapid normal-saline flush.
- If the rhythm persists after 1-2 minutes, give adenosine 12 mg rapid IV/IO with an immediate flush; may repeat the 12 mg dose once if needed. Record the rhythm continuously during administration.
- Do not give adenosine for an irregular rhythm. Contact Medical Control before using a reduced dose for a heart-transplant patient or administration through central venous access.
- For stable atrial fibrillation/flutter causing significant symptoms with SBP of at least 110 mmHg, give diltiazem 0.25 mg/kg IV/IO (maximum 20 mg); if needed, repeat 0.35 mg/kg IV/IO after 15 minutes (maximum 20 mg) when no contraindication is present.
- For prolonged transport, a diltiazem infusion of 5-10 mg/hr requires Medical Control.
- During the approved Claiborne Cardizem shortage only, when diltiazem is unavailable and the same AF/flutter eligibility criteria are met, give metoprolol 2.5-5 mg IV push over 1 minute every 10 minutes; maximum three doses or 15 mg. Reassess blood pressure, heart rate, rhythm, perfusion, respiratory status, and symptoms before every dose.
- Do not independently combine or sequence diltiazem and metoprolol. Contact Medical Control for inadequate response, intolerance, or uncertainty; do not use amiodarone as an AV-nodal substitute for stable AF/flutter with RVR.
- Do not attempt elective rhythm conversion for stable atrial fibrillation/flutter of unknown duration.
- Obtain a post-treatment rhythm strip and 12-lead ECG, reassess complete vital signs, monitor for recurrence, and transport.

Medications

Adenosine

First dose 6 mg rapid IV/IO push with immediate flush; then 12 mg after 1-2 minutes if needed; may repeat the 12 mg dose once

- Paramedic standing order for stable regular narrow-complex tachycardia after unsuccessful modified Valsalva.
- Use the most proximal practical access and record the rhythm continuously during administration.
- Do not use for sinus tachycardia, an irregular rhythm, unstable wide-complex tachycardia, polymorphic tachycardia, or active severe bronchospasm.
- Contact Medical Control before using a reduced dose for a heart-transplant patient or administration through central venous access.

Diltiazem

0.25 mg/kg IV/IO (maximum 20 mg); if needed, 0.35 mg/kg IV/IO after 15 minutes (maximum 20 mg); infusion 5-10 mg/hr

- Paramedic standing order for significantly symptomatic stable atrial fibrillation/flutter when SBP is at least 110 mmHg.
- For prolonged transport, an infusion of 5-10 mg/hr requires Medical Control.
- Avoid with hypotension, acute decompensated heart failure, suspected systolic dysfunction, acute coronary syndrome, pre-excitation, or an undifferentiated wide-complex rhythm.

Metoprolol - Cardizem shortage alternative

2.5-5 mg IV push over 1 minute every 10 minutes; maximum 3 doses or 15 mg

- Paramedic shortage-only alternative for significantly symptomatic stable atrial fibrillation/flutter when diltiazem is unavailable, SBP is at least 110 mmHg, and no AV-nodal-agent contraindication is present.
- Reassess blood pressure, heart rate, rhythm, perfusion, respiratory status, and symptoms before every dose.
- Do not combine or sequence with diltiazem without Medical Control. Avoid with hypotension, bradycardia, advanced heart block, cardiogenic shock, pre-excitation, undifferentiated wide-complex rhythm, acute decompensated heart failure, or suspected systolic dysfunction.

Midazolam

2-5 mg IV/IO; use the smallest effective dose

- Paramedic procedural sedation for synchronized cardioversion when feasible.
- Do not delay cardioversion in an unstable patient.
- Continuous airway, respiratory, SpO_2, cardiac, and blood-pressure monitoring is required.

Warnings / precautions

- Instability must be caused by the tachycardia rather than merely associated with a fast heart rate.
- Do not use adenosine or cardioversion to treat sinus tachycardia.
- Do not perform carotid massage in the prehospital setting.
- Adenosine is contraindicated in active severe bronchospasm and must not be given for an irregular rhythm.
- Do not combine or sequence AV-nodal-blocking medications without an explicit protocol or Medical Control direction.
- In suspected pre-excited atrial fibrillation/flutter, do not administer adenosine, diltiazem, metoprolol, digoxin, or amiodarone.
- Diltiazem may worsen hypotension or decompensated systolic heart failure.
- The metoprolol pathway is a Cardizem-shortage contingency only; it is not a second routine first-line branch and must be withdrawn when diltiazem availability is restored.
- Verify synchronization before every cardioversion attempt; many devices require synchronization to be reactivated after each shock.
- Sedation must not delay cardioversion when instability is attributable to the rhythm.

Clinical pearls

- A tachyarrhythmia causing instability is often at least 150/min, but rate alone does not determine treatment.
- A properly performed modified Valsalva is more effective than a simple seated Valsalva maneuver.
- Adenosine may transiently reveal atrial activity without terminating atrial fibrillation or flutter; it is not routine therapy for an irregular rhythm.
- Capture the rhythm before, during, and after adenosine or cardioversion whenever feasible.
- Stable atrial fibrillation/flutter of unknown duration should receive rate control rather than elective prehospital rhythm conversion.

Special populations | Heart transplant or central venous access

- Adenosine may produce an exaggerated response in a heart-transplant patient or when administered centrally.
- Contact Medical Control before using a reduced dose; do not automatically give the standard 6 mg dose through central access.

Special populations | Pre-excited atrial fibrillation/flutter

- An irregular rapid rhythm with variable QRS morphology or known Wolff-Parkinson-White may represent pre-excitation.
- Avoid AV-nodal blockers and amiodarone; perform synchronized cardioversion if unstable and contact Medical Control early if stable.

Related pathways / contacts

AC-04 ACS / STEMI - Open for ischemic chest discomfort or suspected acute coronary syndrome.

AC-05 Acute Pulmonary Edema - Open when acute heart failure or pulmonary edema is present.

AC-07 Monomorphic Wide-Complex Tachycardia - Open when the QRS is wide and the rhythm is monomorphic.

AC-08 Polymorphic Wide-Complex Tachycardia - Open for polymorphic or torsades-pattern tachycardia.

AM-05 Hypotension / Shock - Open for persistent hypotension or shock.

AC-03 Adult Cardiac Arrest - Open immediately if the patient becomes pulseless.

Source and references

American Heart Association. 2025 Adult Tachyarrhythmia With a Pulse Algorithm. | American Heart Association. 2025 Electrical Cardioversion Algorithm. | American Heart Association. 2025 Guidelines for CPR and ECC: Adult Advanced Life Support. | Tennessee Emergency Medical Services Board. Tennessee EMS Protocol Guidelines, September 2025. | Claiborne EMS medication formulary and related adult cardiac protocols. | American College of Cardiology/American Heart Association/American College of Clinical Pharmacy/Heart Rhythm Society. 2023 Guideline for the Diagnosis and Management of Atrial Fibrillation.

Revision: October 16, 2026 | Last verified: October 16, 2026